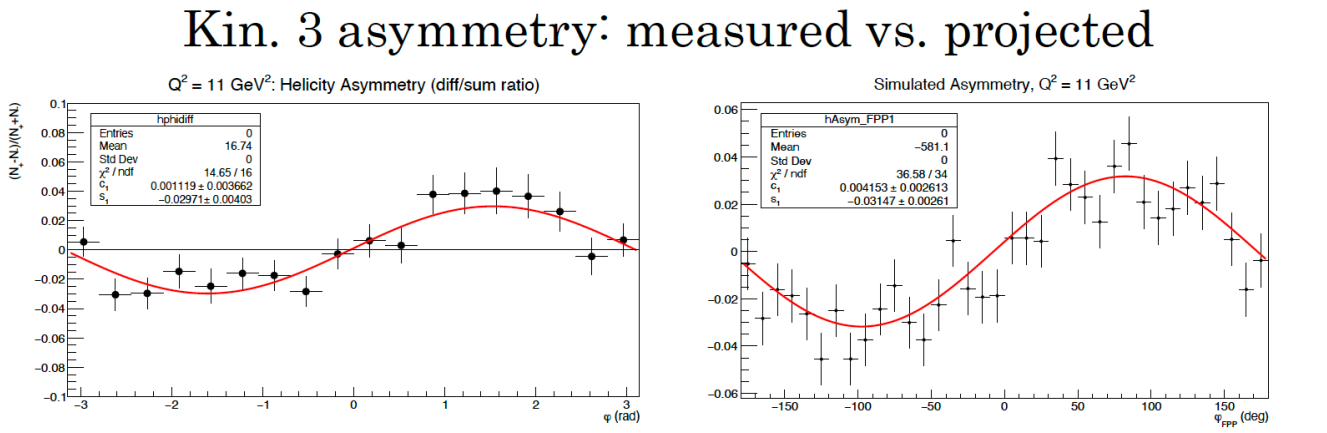
Subject: RE: [EXTERNAL] *Reminder* FPP run group meeting
Date: Tuesday, November 10, 2026 at 11:35:29 AM Eastern Standard Time
From: Andrew Puckett
To: Ed Brash
Attachments: image.png, image.png, image.png

PS—Sorry, I realized I accidentally clicked “send” on that before finishing bullet point 5) below.

Just wanted to remind you of the comparison I showed on slide 36 of my collaboration meeting talk way back in March.

The very early, preliminary, low-precision asymmetry extraction from the SBS data is consistent with the simulation prediction, and the simulation prediction is based on the GEp-III analyzing power parametrization. So it looks like the analyzing power at Q^2 of 11 GeV^2 is roughly in line with expectations.

Cheers,
Andrew



- Left: Real data asymmetry from ~140k elastic ep events with scattering angles in the useful range $0.07 \leq p_T(GeV) \leq 1.0$ (this represents approximately $1/4^{\text{th}}$ of “worst case” statistics accounting for polarimeter efficiency
- Right: Asymmetry from ~440k simulated events with scattering angles in the useful range (note this is not supposed to represent full proposal stats!)
- Consistency of measured and simulated asymmetries suggests analyzing power is good and consistent with (extrapolated) GEp-III parametrization

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From: Andrew Puckett <puckett@jlab.org>
Date: Tuesday, November 10, 2026 at 11:30 AM
To: Ed Brash <brash@jlab.org>
Subject: RE: [EXTERNAL] *Reminder* FPP run group meeting

Hi Ed,

Sorry, life intervened and it took me longer to get back to you on this than I anticipated. I started drafting this on Friday.

That's a lot of progress in a short timeframe. Quite interesting and will take some time to digest. I suspect your FOM estimate is probably considerably more accurate than mine, but it is also more than a factor of two higher than mine.

I believe I can obtain an "apples-to-apples" comparison between the number quoted in my spreadsheet and the number you described as "FOM per incident proton", by simply dividing my number by the total number of projected events. Under that assumption, my quick-and-dirty spreadsheet calculation of FOM gives 3.6 x 10^(-4) per incident proton, compared to your 7.6e-4.

That means the projected FF ratio uncertainty in my spreadsheet should be reduced by a factor of sqrt(2) (all else equal) if your numbers are more accurate, as I suspect they are (at least more accurate than mine). However, see caveats below.

My estimate of an "average" Ay of 0.05 is based on a simple linear extrapolation from the numbers published in Table VII of our 2017 PRC paper:
https://inspirehep.net/literature/1613323

(as well as our "Technical Supplement"): https://inspirehep.net/literature/1611577

But it's not exactly clear to me how the "average" Ay in that table was defined. The answer is, of course, very important to the implications for FOM.

Moreover, the difference between your projection and mine might be related to the difference between weighting events by the analyzing power in the likelihood estimators to optimize the statistical precision of the extraction, versus just throwing all the events within the accepted p_T range into one asymmetry plot and trying to pull out the FFR from that. I thought my projections looked too pessimistic, even compared to what I was doing to estimate "best-case" and "worst-case" scenarios for the SBS analysis.

A few questions and comments:

1) The single-track/multi-track dichotomy. As you may remember, the GEp-III analyzing

power results as quoted in the PRC paper are for single-track events only. If your quoted “efficiency” of ~26% includes multi-track events, it is probably somewhat of an overestimate. Indeed, the quoted “efficiencies” in Table VI of the PRC paper are for the “double-FPP” layout and include only single-track events. For a single CH2 analyzer of the thickness used in GEp-III, the single-track efficiency at high Q^2 might be even less than my conservative 20% assumption. For example, at the highest Q^2 of GEp-III, the effective “FPP1” efficiency was “only” 15%. However, I suspect with 12-plane tracking and better resolution of the left-right ambiguity and better overall tracking efficiency, what we get in practice will be considerably higher, probably closer to 20%. The numbers in Tab. VI of the PRC paper are based on events passing ALL FPP event selection criteria, presumably the ones described in the following table from the “technical supplement” NIM paper. With the improved accuracy, efficiency, and resolution of 12-plane tracking, I am sure we will have a higher effective efficiency than we achieved in practice for the GEp-III analysis. But 26% seems a bit on the high side if we restrict the analysis to single-track events (as we should when using the GEp-III analyzing power parametrization).

Table 1: FPP event selection criteria as a function of Q^2. Only single-track events passing the "cone test" were included in the analysis. No explicit ϑ cuts were applied. Instead, the ϑ ranges shown are the effective ranges resulting from the p_T cuts. The same criteria were applied to all three ϵ values at $Q^2 = 2.5$ GeV^2. s_{close} and z_{close} are defined, respectively, as the distance of closest approach between the incident and scattered tracks, and the z-coordinate of the point of closest approach between incident and scattered tracks, with $z = 0$ at the HMS focal plane. See text for details.

Q^2 (GeV^2)	2.5	5.2	6.8	8.5
p_T^{min} (GeV/c)	0.06	0.05	0.05	0.05
p_T^{max} (GeV/c)	1.2	1.5	1.5	1.5
FPP1 $\vartheta_{min}^{eff}(^\circ)$	1.71	0.81	0.65	0.53
FPP1 $\vartheta_{max}^{eff}(^\circ)$	36.7	25.1	19.9	16.3
FPP2 $\vartheta_{min}^{eff}(^\circ)$	1.82	0.84	0.67	0.55
FPP2 $\vartheta_{max}^{eff}(^\circ)$	39.5	26.0	20.4	16.6
FPP1 s_{close}^{max} (cm)	2.2	1.7	1.4	1.2
FPP2 s_{close}^{max} (cm)	6.5	5.1	4.1	3.3
FPP1 z_{close}^{min} (cm)	108	108	108	108
FPP1 z_{close}^{max} (cm)	168	168	168	168
FPP2 z_{close}^{min} (cm)	207	207	207	207
FPP2 z_{close}^{max} (cm)	267	267	267	267

2) How exactly are you implementing the GEp-III Ay parametrization? In the code I use to estimate SBS FOM, it appears I use a weighted average of the "b" parameters from the three highest Q^2 points from Table VII of the PRC paper, together with the linear fit to the Aymax values to extrapolate/interpolate Ay to any given (p,p_T) value on an event-by-event basis:

fits to the GEp-II data shown in Fig. 15 only include the three highest Q^2 points, which used the same analyzer thickness. It is also worth noting that the Dubna measurements [54] different from zero. The experimentally realized effective analyzing power for the Hall C double-FPP is substantially greater than that of the GEp-II or Dubna polarimeters at similar

TABLE VII. $A_y(p_T)$ fit results. Fit parametrization is as in Eq. (33). Uncertainties in fit parameters are statistical only. Parameters with no uncertainties are fixed at the quoted values. $A_y^{\max}$ is the maximum value of A_y occurring at $p_T = p_T^{\max}$. Uncertainties in derived quantities $A_y^{\max}$ and $p_T^{\max}$ are computed from the full covariance matrix of the fit result and the gradient of the fit function with respect to the parameters evaluated at the maximum. $\bar{A}_y$ values are for $0.06 \leqslant p_T$ (GeV)$\leqslant 1.2$.

Q^2 (GeV^2)	2.5 ($\langle\epsilon\rangle = 0.153$)	5.2	6.8	8.5
N	0.44 ± 0.02	0.48 ± 0.03	0.44 ± 0.02	0.35 ± 0.01
α	0.48 ± 0.02	1	1	1
β	1.89 ± 0.05	2	2	2
b	2.05 ± 0.03	2.4 ± 0.2	2.8 ± 0.2	2.3 ± 0.1
p_T^0 (GeV)	0.053 ± 0.002	0	0	0
$A_y^{\max}$	0.2027 ± 0.0006	0.134 ± 0.005	0.114 ± 0.004	0.098 ± 0.003
$\bar{A}_y$	0.1471 ± 0.0003	0.085 ± 0.003	0.073 ± 0.003	0.061 ± 0.002
$p_T^{\max}$ (GeV)	0.38 ± 0.03	0.46 ± 0.09	0.42 ± 0.08	0.47 ± 0.08

3) I did a bit more digging into how I did the SBS FOM estimates to refresh my memory. I was able to reproduce my ~9M event “best case” projection for Kin. 3. Basically, it starts with the g4sbs MC simulation, runs everything through digitization and reconstruction, and then applies cuts on ECAL and HCAL energy to mimic the trigger, dx and dy and “dpp” to mimic the exclusivity cuts, and requires a track in both FT and FPP, with the “sclose” in the FPP less than 3 mm (but no cut on the FPP polar angle yet). In fact, the projected number of events passing all those cuts is close to 12M, but then I apply a 70% overall “efficiency” factor for what I think is a fairly optimistic best case for what we might eventually achieve. That’s how I arrive at ~9M. However, since this cut already enforces the requirement of a track in the FPP and a small distance of closest approach, and a large energy deposit in HCAL, the “efficiency” factor relative to this number is actually closer to 33%, for a total of roughly 2.9M projected useful events for polarimetry. That number is considerably less than the roughly 5.1M events that my “quick and dirty” FOM estimator uses to project a FF ratio uncertainty of ~0.1 (absolute stat. uncertainty in mu_p GE/GM). The difference between the two estimates mainly reflects the “ideal” reconstruction efficiency (what we get from background-free simulation + digi + recon) and the effect of tight exclusivity cuts. If I just do a naive sqrt(N) scaling from 5.1M to 2.9M useful events, I obtain a FF ratio uncertainty of about 0.13, which matches what I’ve been showing for my “best case” estimate. HOWEVER, and this is important for the comparison to Hall C, the yield estimate is based on a pure Born-approximation event generator, not a fully-radiated event generator like in SIMC. Therefore, the projected number of useful events for the SBS measurement is probably at least ~10-15% too optimistic (if not more) at this Q^2, making the Hall C comparison more attractive (for Hall C).
4) Regarding the one-analyzer tracking strategy, your proposed approach seems superficially reasonable. I assume that for high-Q^2 measurements we will prefer to increase the (single-analyzer) CH_2 thickness to perhaps 75-90 cm (that number is based on previous FOM optimization studies for the SBS measurement), and

rearrange the tracking layout a bit as needed. Note that at some point in the distant past I had already demonstrated a working 12-plane tracking algorithm in the FPP chambers using straight-through data. I can probably dig up my private version of the Hall C fortran code that contains that algorithm. It certainly used a “brute force” combinatorial approach, although obviously bad wire combinations were ruled out from the fit to wire positions only, and then only the wire combinations representing potential straight-line tracks based on the fits to wire positions only then have all left-right combinations tested for the chi2 of a linear fit to wire positions +/- drift distances. I remember this code running pretty slowly compared to the six-plane version, for obvious reasons, but I bet it would still run much faster than the SBS code, especially on the more modern batch farm hardware. In any case, I have no concerns regarding the speed of tracking in the Hall C/HMS context.

5) Finally, regarding the question about how well-anchored our analyzing power extrapolation is, I fully agree that since the Hall C kinematics are basically identical to the Hall A kinematics, the uncertainties of the analyzing power estimates are basically 100% correlated. But I remind you of the plot I showed in

Okay, so I think that’s about all I probably had to say about this subject. But pending the answer to the single-track/multi-track question, I think we go with your FPP FOM estimate. I’d love to chat more about your FPP simulation(s) on Zoom in the near future, but there’s no special urgency.

I did, however, want to close the loop on this email I promised you on Friday.

Cheers,
Andrew

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From: Ed Brash <brash@jlab.org>
Date: Friday, November 6, 2026 at 1:00 PM
To: Andrew Puckett <puckett@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

Hi Andrew,

I have updated the short GEp-4 feasibility presentation that I showed on Wednesday to include progress that I have made on the Hall C FPP GEANT4 simulations over the last couple of days. I have tried to make the progression and assumptions explicit.

In brief, I ported the actual Hall C FPP chamber geometry and wire mapping from the legacy GEp-III GEANT3 simulation into GEANT4. The current reconstruction uses the twelve U/X/V planes, unsigned drift distances, explicit enumeration of the left/right solutions, simultaneous straight-line fits in x(z) and y(z), and minimum-chi-squared track selection. From the reconstructed HMS and FPP tracks, I calculate the usual theta_polar, phi_az, z_close, and s_close and apply the cone test.

The current analyzer configuration that I used is for a single legacy 22-inch CH2 analyzer - this is, to the best of my knowledge, the maximum analyzer thickness that we can fit in the existing FPP frame (with about 1.5 inches of clearance on either side). Below 12 degrees, all four FPP chambers are used to constrain the left/right ambiguity. Above 12 degrees, FPP1 and FPP2 alone are used, which avoids the downstream cone-test acceptance loss where the larger scattering angle already makes the ambiguity less troublesome. I understand that for Gep-4, there is essentially no figure of merit above about 10 degrees, but I was also thinking about the nuclear FF experiment at the same time, where larger angle scatterings are more prominent and need to be recovered. So, the point is that I just implemented this 2 or 4 chamber algorithm from the outset.

So, the present tracking that I have is real wire-level tracking, but the detector inputs remain idealized: hits are truth-associated, drift distances are exact, and chamber inefficiency, resolution, noise, dead regions, and multi-track pattern recognition have not yet been included. Still, I think that this are minor effects in general, and we hopefully don't need to worry about them too much for a proposal-level feasibility study.

For the FOM calculation, I used the GEp-III/GEp-2gamma empirical analyzing-power parameterization that was subsequently used in SBS planning:

FOM per incident proton = epsilon_FPP x (P_e A_y)^2

and

sigma_component = sqrt[2/(N x FOM)].

For the current 6.7 GeV/c calculation, the useful-pT efficiency is 26.264%, the effective analyzing power is 0.06311, and the beam-polarization-weighted FOM is 7.558 x 10^-4 per incident proton. Applying that to the provisional 7.5–7.7 million-event sample gives a projected single-component uncertainty of approximately 0.0188–0.0185. I want to emphasize that I just did the back-of-the-envelope calculation for SBS using effective numbers, but for Hall C, I ran the actual GEANT4 simulation and extracted the integrated

Ay^2 weighted FOM.

The quoted effective A_y = 0.06311 deserves some explanation. It is the A_y-squared-weighted RMS over the accepted useful-pT events, not the peak or an ordinary average. The parameterization is extrapolated to 6.7 GeV/c and has not been independently validated by the completed SBS analysis to the best of my knowledge at this point, although you might have some preliminary numbers for this from some your analyses - I am not sure. However, because the GEp-4 and GEp-5 proton kinematics are so similar, the uncertainty in the absolute analyzing power should be strongly correlated between them. If the true effective A_y is lower, it should reduce the statistical power of both experiments by approximately the same A_y-squared factor. So, clearly it is important for the absolute GEp-4 uncertainty, but less problematic for the relative GEp-4 versus GEp-5 feasibility comparison.

The presentation also retains the earlier data-anchored yield, HMS-trigger, and NPS-rate studies. Since Wednesday's meeting, the NPS background study has also achieved some additional closure. Basically, I decided to separate target-background generation from calorimeter transport: GEANT4 first records the energetic charged and neutral particles produced by beam electrons in the LH2 target, and those particles are then transported independently through the NPS GEANT4 model. For run 6887, the combined inclusive-electron and electronuclear contributions now reproduce the measured NPS trigger-rate scale and threshold dependence reasonably well over the region of interest. The agreement is not perfect, particularly at the highest thresholds, but it is substantially better than the inclusive-electron calculation alone and supports the conclusion that photon-induced showers supply much of the previously unexplained rate. The sweeper-magnet effect appears small in this comparison. I now have considerably more confidence that this framework contains the dominant background physics needed for a GEp-4 NPS-rate projection, although there is still some additional work to be done. But, as you said, even with a super-conservative (1 MHz) NPS background rate, we are probably still fine.

I would be very interested in your reaction, particularly concerning the one-analyzer tracking strategy and the analyzing-power convention.

Cheers,

E.

Dr. Edward J. Brash
(He/him)
Professor of Physics - Christopher Newport University
Staff Scientist - Thomas Jefferson National Accelerator Facility
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Office: 757-594-7451
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From: Andrew Puckett <puckett@jlab.org>

Date: Thursday, November 5, 2026 at 6:28 PM
To: Ed Brash <brash@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

Ah yes, I may have missed that detail in the previous email. So it seems like the to-do list is now down to interfacing the SIMC output with some kind of polarimeter simulation to obtain a more accurate estimate of the polarimeter FOM with an optimized layout, and improving our estimate of the calorimeter rates and the impact of calorimeter response on elastic efficiency/yield. I am now more or less fully convinced that the proposal is technically feasible. My initial estimate of polarimeter FOM at 11 GeV^2 is a trifle disappointing, but the projection still more or less matches the SBS "best case" scenario for equivalent beam time.

Andrew

puckett.physics.uconn.edu

> On Nov 5, 2026, at 6:14 PM, Ed Brash <brash@jlab.org> wrote:
>
> Hi Andrew,
>
> Okay, I see now what the issue was. The 1.8 factor that I quoted before was coming because of SHMS acceptance, not HMS acceptance (for the most part).
>
> The approximately 10.7-million-event projection is unchanged though. That calculation used the complete NPS/SHMS acceptance ratio, so the problem was my interpretation of the intermediate 1.8 factor, not the final projected yield.
>
> I independently confirmed the same ratios (for HMS acceptance) that you got: the 20 cm/10 cm ratio is 1.85, while the 30 cm/10 cm ratio is 2.57.
>
> Sorry for the confusion.
>
> Best,
> E.
>
> Dr. Edward J. Brash
> (He/him)
> Professor of Physics - Christopher Newport University

Staff Scientist - Thomas Jefferson National Accelerator Facility
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From: Andrew Puckett <puckett@jlab.org>
Date: Thursday, November 5, 2026 at 5:05 PM
To: Ed Brash <brash@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

PPPS—for the Q^2 = 11 kinematics, the yield ratios are:

20 cm/ 10 cm: 1.85
30 cm/ 10 cm: 2.55

The (xsec-weighted) vertex z distribution is more skewed toward the downstream end of the target for Q^2 = 11 (I suppose we shouldn't be surprised by that, given the correlation between vertex z and Q^2).

Interestingly enough, the 20 cm/10 cm yield ratio is about the same at 18.8 degrees as at 24.7 degrees, but the 30 cm/10 cm yield ratio is greater (but not by much) at the more forward angle as I would naively expect.

From: Andrew Puckett <puckett@jlab.org>
Date: Thursday, November 5, 2026 at 4:57 PM
To: Ed Brash <brash@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

PPS—just looking at the (cross section weighted) vertex z distribution from a simulation of Q^2 = 6 at a central angle of 24.66 degrees seems to show good acceptance along the entire target length. So I find it very hard to believe that increasing target length by a factor of 3 "only" increases the accepted elastic rate by 1.8X. Again, this is for Q^2 = 6, with a larger HMS central angle than our high Q^2 point, but for this Q^2 I find that if I take the "vertex" value of z (not smeared for optics and detector resolution) and just calculate the integral of the distribution within +/-5, 10, 15 cm, I get:

Elastic yield (20 cm / 10 cm) = 1.86
Elastic yield (30 cm / 10 cm) = 2.48

It's clear that we'd start to see diminishing returns beyond 30 cm, but at more forward angles I'll bet the HMS can "see" even more useful target length. I

wonder if we ought to think about a 40-cm target (clearly more difficult from a design and cryo standpoint, but this will run in the “post MOLLER” era.

Andrew
<image.png>

From: Andrew Puckett <puckett@jlab.org>
Date: Thursday, November 5, 2026 at 4:41 PM
To: Ed Brash <brash@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

PS—You had mentioned earlier that the elastic rate only increased by a factor of 1.8 when going from a 10-cm to a 30-cm target in some of your test/benchmarking kinematics. If that’s really true, then it’s possible that a 30-cm target would, in fact, be overkill. We should look at a 20-cm target as well, because if the HMS extended-target acceptance can’t make efficient use of a full 30-cm target, there’s not much point in using a longer target than we can efficiently use, when considering the fact that the calorimeter would effectively “see” the entire target length.

From: Andrew Puckett <puckett@jlab.org>
Date: Thursday, November 5, 2026 at 4:07 PM
To: Ed Brash <brash@jlab.org>
Subject: Re: [EXTERNAL] *Reminder* FPP run group meeting

Hi Ed,

I just wanted to share with you a bit of progress on a few “quick and dirty” SIMC rate estimations and uncertainty projections I’ve been doing, thinking ahead to a full proposal. I attach a spreadsheet with the calculations.

The columns are hopefully self-explanatory. For the main GEP-4 proposal effort, I’ve depicted a total of 100 PAC days divided among four Q^2 points at 4, 6, 8, and 11 GeV^2.

I ran SIMC for each of these kinematics, always assuming NPS calorimeter at 3 meters for the electron arm. I checked the generation limits and they are sufficiently wide. All four of these Q^2 points are chosen with epsilon ~0.7, which leads to similar Jacobian and solid-angle matching across all four points. Indeed, checking the envelope of elastic events on NPS, the calorimeter

appears to comfortably cover the needed solid angle to more than match that of the HMS at 3 meter distance, although the distance could be optimized a bit in a concrete run plan. The required central angles of the calorimeter are similar enough between Q^2 = 8 and 11 GeV^2 that the calorimeter could be left in the same position for both of those points.

The rate estimate includes loose exclusivity cuts (in fact, significantly looser than we would apply in offline analysis) of W < 1.1, |Em|<0.2, Pm<0.2 (which basically just excludes events with very large radiative energy losses), so the implied rates are slight overestimates relative to what the final analysis yields might be (also, I don't include any kind of overall target boiling or livetime/efficiency factors yet). And, since I don't have a calorimeter simulation or any calorimeter smearing yet, these estimates do NOT include the effect of calorimeter response on the efficiency.

For the polarimeter FOM estimation, I've assumed a constant "efficiency" factor of 20% (for scattering in the useful range of angles in the CH2), and estimated the corresponding average analyzing power for the events used for polarimetry by interpolating/extrapolating from GEp-III/2gamma data. I've then taken the "FOM" to be:

FOM = Num events (column V) x polarimeter efficiency x (0.85 * estimated average Ay)^2

(The 0.85 accounts for beam polarization).

From this quantity, we can estimate the uncertainties of each of the measured polarization components as:

Delta_stat (Pyfpp or Pxfpp) = sqrt(2 / FOM)

The 20% efficiency estimate for the polarimeter should probably be regarded as a somewhat conservative "worst case", particularly if we increase the CH2 thickness for higher-Q^2 measurements.

The uncertainty in Pt at the target, to a good approximation, is the same as the uncertainty in Pyfpp, while the uncertainty of PL at the target is, to a less good approximation for the HMS, equal to the uncertainty in Pxfpp/ sin(chi), where chi is the (central) precession angle.

In fact, in the HMS context, increasing the PL uncertainty by 1/sin(chi) pretty drastically overestimates the final uncertainty of PL due to the large chi acceptance and the weighting of events by spin transport matrix elements in